Aashwat Singh

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EDUCATION

Bachelor of Science in Computer Science | University of Texas at Dallas, Richardson, TX *Expected May 2028*
Relevant Coursework: Computer Architecture & Assembly Language (MIPS), Digital Logic & Circuits, Digital Systems Lab, Data Structures & Algorithms, Programming Paradigms, Software Engineering, Discrete Math, Linear Algebra

PROFILE

Computer Science student seeking a Summer 2027 firmware / embedded software internship. Writes low-level code in MIPS assembly and C/C++, designs and debugs sequential logic on real TTL hardware, and verifies digital designs in Verilog. Comfortable working at the register and memory-layout level, reading chip pinouts, and backing code with tests and tooling carried over from ML and backend projects.

TECHNICAL SKILLS

Languages: C, C++, MIPS Assembly, Verilog, Python, Java, Rust, SQL, TypeScript
Embedded & Hardware: MARS (MIPS simulator), Logisim Evolution, Verilog testbenches, 74LS-series TTL logic, breadboard prototyping, FSM design, fixed-point & two's-complement arithmetic
Tools & Testing: Git, GitHub Actions CI, pytest, property-based testing, Docker
Software & ML: FastAPI, Node.js, PostgreSQL, React, Pandas, Scikit-Learn, OpenCV, LLM integration (Claude API)

PROJECTS

Black Hole: Strategy Game in MIPS Assembly | *MIPS, MARS* *Spring 2026*

- Wrote a 4-module, ~330-line turn-based game in pure MIPS assembly with no macros, linking modules through **.globl** symbols and a **jal / jr $ra** subroutine calling convention with arguments in $a0 and results in $v0.
- Packed the 21-cell triangular board into a flat 21-byte array, encoding piece owner and value in a single byte (1–10 player, 11–20 computer), and resolved 2D pyramid neighbors through a 21×6 byte lookup table in the data segment.
- Implemented bounds and occupied-cell input validation, a random-move opponent with collision retry (syscall 42), an adjacency-scoring win check, and an ASCII pyramid renderer driven by nested row/column loops.

4-Bit Accumulator ALU | *74LS74, 74LS283, breadboard* *Fall 2026*

- Built a 4-bit accumulator (A ← A + B) from two 74LS74 dual D flip-flops and a 74LS283 adder, feeding each sum bit back to its flip-flop's D input, with shared clock/clear lines and carry-out driven to an LED.
- Characterized edge-triggered D (74LS74) and JK (74LS107) flip-flops on hardware, confirming asynchronous active-low PRESET/CLEAR, JK toggle mode, and the invalid PRESET = CLEAR = 0 state.

Predictive NBA Analytics Platform | *Python, Pandas, Scikit-Learn* *July 2026 - August 2026*

- Built an end-to-end Python pipeline modeling 700+ NBA stat records to predict multi-variable performance metrics, isolating train/test sets to eliminate structural data leakage.
- Benchmarked and tuned Linear Regression against a Random Forest Regressor, reaching a verified 91.8% accuracy.

WashBuddy Full-Stack Application | *React, Node.js, MongoDB, Gemini* *September 2025*

- Integrated a 4-person team's AI laundry-sorting app, wiring the Gemini frontend to a MongoDB API and configuring backend classification so clothes grouped correctly; won Best Hack for Beginners at HackWesTX VI.

PROFESSIONAL EXPERIENCE

AI / Computer Vision Extern — Outamation *Jan 2026 - March 2026*

- Built a Python ML pipeline (PyMuPDF, OpenCV, LlamaIndex) extracting structured financial data from mortgage documents, achieving 95% extraction accuracy and reducing manual review time by 75%.

AI/ML Research Trainee — CRA *July 2025 - August 2025*

- Applied OCR and computer vision to documents that arrive as scans and photographs rather than clean digital files, in an intensive program covering supervised learning, NLP, and computer vision.

RESEARCH & LEADERSHIP

Lead Researcher — Web3 Acceleration Association *Jan 2025 - Dec 2025*

- Led a research team analyzing 100+ blockchain transactions to evaluate a decentralized token's transaction security; co-authored and presented the resulting paper to a non-engineering audience.

Student Researcher — AIMD *September 2026 - Present*

- Developing Project Wall-E, an edge-AI triage proposal for rural primary care, for the UT Dallas AI in Medicine Case Competition (HIMSS27).